

XUEJIAN(JACOB) SHEN

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Massachusetts Institute of Technology
77 Massachusetts Ave, Cambridge, MA 02139, USA

EDUCATION

California Institute of Technology Ph.D., Physics; Advisor: Prof. Philip Hopkins Thesis: Cosmic structure and galaxy formation in alternative dark matter	Sept 2018 - June 2023
Peking University B.S., Physics	Sept 2014 - June 2018

EMPLOYMENT

Postdoctoral Researcher Kavli Institute for Astrophysics and Space Research Massachusetts Institute of Technology	Oct 2023 - present
SubMIT Project Team Department of Physics, Massachusetts Institute of Technology	Oct 2023 - present
Graduate Teaching & Research Assistant California Institute of Technology	Jan 2019 - June 2023

RESEARCH INTERESTS

Cosmological simulations of galaxy formation; The nature of dark matter; Formation and evolution of galaxies and supermassive black holes at high redshift
UAT Keywords: galaxy formation (595) – galaxy evolution (594) – dark matter (353)

SUMMARY OF PUBLICATIONS

Total publications: **81**
First-author publications: **14**; From supervised students: **9**
Metrics: **>2900 citations**, **>1100 first/corresponding-author citations**, **h-index: 29**
See my [NASA ADS bibliography](#) for full information

AWARDS & FELLOWSHIPS

Outstanding UROP Mentor Award	Massachusetts Institute of Technology, 2026
ITC Fellowship	CfA, Harvard University, 2025
Neil Gehrels Prize Fellowship (declined)	University of Maryland, 2023
KIPAC Fellowship (declined)	Stanford University, 2023
TCCAP Fellowship (declined)	University of Texas at Austin, 2023
James A. Cullen Memorial Fellowship	California Institute of Technology, 2022
Honored Graduate	Peking University, 2018
Robin Li Fellowship	Peking University, 2017
Meritorious Award	Mathematical Contest in Modeling, 2017
Weiming Scholar Fellowship	Peking University, 2015,2016
HaiLiang Fellowship	Peking University, 2016
GuangHua Fellowship	Peking University, 2015

RESEARCH GRANTS

As Principal Investigator (PI)		
JWST-AR-Cycle3 Theory (04814)	\$329597.05	10/2024 – 10/2026
As Co-Investigator		
JWST-GO-Cycle4 (07404, PI Naidu)	49.3 hours	10/2025 – 10/2027
SuperMUC-NG at LRZ (PI Puchwein)	33M core hours	09/2025 – 09/2027
DiRAC THESAN-MultiZoom (PI McClymont)	18.4M core hours	01/2026 – 01/2028
THESAN-CGM refinement (PI Pruto)	5.2M core hours	01/2026 – 01/2028

TEACHING & ADVISING

Caltech Graduate Teaching Assistant, Computational Physics Lab Ph20/21/22
Lectures, MIT Physics 8.902 (2023,2024)

Undergraduate & Master Students:

Gabriel Aguiar (Caltech, 2021-2022), Eitan Rapaport (Caltech, 2022-2023), Evan Erickson (MIT, 2023-), Yongao Hu (MIT, 2024), Hui Wang (MIT, 2023-2024), Vinh Tran (MIT, 2023-), Aidan Leonard (MIT, 2023-2024), Zihao Wang (MIT/PKU, 2023-), Charline Shen (Harvard, 2023-2025), Qiaorong Yu (Oxford, 2025), Xiaoqing Sun (MIT, 2024), Aliaksandr Melnichenka (Berea College, 2025-), Hyun Song (Harvard, 2026-), Ye He (NJU, 2026-), Jiahui Xie (Caltech/Harvard, 2026-), Chuiyang Kong (Brown/Harvard, 2026-)

Graduate Students: Rongrong Liu (Harvard, 2025-), Darkhan Shadykul (MIT, 2026-)

SYNERGISTIC ACTIVITIES

Professional Services

- Journal Referee for Monthly Notices of the Royal Astronomical Society (MNRAS, since 2020), MNRAS Letters (since 2023), The Astrophysical Journal (ApJ, since 2022), ApJ Letters (since 2022), Astronomy & Astrophysics (since 2025), Scientific Reports (Since 2023), Physics Review Letters (PRL, since 2024), Physics Review D (PRD, since 2023)
- Oscii Bascii (& Grad student representative) of Theoretical AstroPhysics Including Relativity

- and Cosmology (TAPIR, 2019-2022), Caltech
- Local Organizing Committee (LOC) for the Galaxy Formation and Evolution in Southern California (GalFRESKA) workshop (2022)
 - MIT Physics Basic Computing Services “SubMIT” Project Team Member (since 2023)
 - Organizer for MIT Physics Basic Computing Services User Meetings (since 2023)

Outreach

- Speaker at Caltech Stargazing Lecture Series (2023)
- Organizer of Stargazing and Outreach Activities (2015-2018), Peking University

SELECTED TALKS

Invited Talks

- Learning the Universe at Cosmic Frontier Penn State University, 04/2026
- Learning the Universe at Cosmic Frontier Westlake University, 01/2026
- Galaxy formation models at the cosmic dawn: Insights from JWST Shanghai Jiaotong University, 01/2026
- Learning the Universe at Cosmic Frontier Nanjing University, 12/2025
- Learning the Universe at Cosmic Frontier Fudan University, 12/2025
- Cosmic Dawn as a Laboratory for New Physics Shanghai Jiaotong University, 12/2025
- Cosmic Dawn as a Laboratory for New Physics HEP seminar, Harvard University, 11/2025
- Cosmic Dawn as a portal to New Physics Boston University, 10/2025
- Cosmic Dawn as a portal to New Physics Nanjing University, 09/2025
- Cosmic Dawn as a portal to New Physics Peking University, 09/2025
- Cosmic Dawn as a portal to New Physics Tsinghua University, 09/2025
- Alternative dark matter in the THESAN simulations DREAMS collaboration meeting, CCA, 05/2025
- Challenges to galaxy formation models at the cosmic frontier Yale University, 04/2025
- Challenges to galaxy formation models at the cosmic frontier CCA, Flatiron Institute, 10/2024
- Challenges to galaxy formation models at the cosmic frontier Zhejiang University, 09/2024
- Challenges to galaxy formation models at the cosmic frontier Univ. of Texas Austin, 08/2024
- Challenges to galaxy formation models at the cosmic frontier Univ. of Cambridge, 06/2024
- The implication of UV variability for the bright galaxy abundance at cosmic dawn Tsinghua University, 09/2023
- The implication of UV variability for the bright galaxy abundance at cosmic dawn KIAA seminar, Peking Univ., 09/2023
- Dark Matter: Elusive fibers of the Universe Stargazing Lecture, Caltech, 05/2023
- New aspects about dark matter models beyond CDM KIPAC tea talk, Stanford, 02/2023
- Alternative dark matter and structure formation Obs. Cosmo. seminar, Caltech, 11/2021

Contributed Talks

- An Introduction to the LUMINA project Journal Club, MIT. 04/2026
- The cosmic rush hour in a universe with early dark energy Tokyo, 03/2026
- The cosmic rush hour in a universe with early dark energy Aspen, 03/2026
- The cosmic rush hour in a universe with early dark energy ITC Luncheon, 02/2026
- The cosmic rush hour in a universe with early dark energy Napa Valley, 01/2026
- Cosmic Dawn as a portal to New Physics FOLIAGE meeting, 11/2025
- Quasar luminosity functions and clustering in THESAN-XL Thesan workshop, Harvard CfA, 08/2025

- The rich astrophysical phenomenology in a dissipative dark sector SIDM workshop, Valencia, 06/2025
- Flash talk: Supermassive black hole seeds from dark matter CFC meeting, Austin, 05/2025
- Basic Computing Service Review: OpenMPI and Data Transfer MIT, 05/2025
- First results from the THESAN-ZOOM simulations Journal Club, MIT, 03/2025
- Star-formation efficiencies in high-redshift galaxies learned from THESAN-ZOOM Harvard CfA, 03/2025
- Early dark energy as a unified solution to the Hubble tension and early galaxy puzzles raised by JWST KITP conference, 08/2024
- Insights on efficiency and variability of star-formation from radiation-hydro simulations Santa Cruz Galaxy Formation Workshop, 08/2024
- Basic Computing Service Review: Customizations and Community Engagement MIT, 08/2024
- Challenges to galaxy formation models at the cosmic frontier Journal Club, MIT, 04/2024
- Challenges to galaxy formation models at high redshifts University of Vienna, 02/2024
- Galaxy sizes during the epoch of reionization “Building Galaxies from Scratch” Harvard CfA, 02/2024
- The implication of UV variability for the bright galaxy abundance at cosmic dawn ITC luncheon, Harvard CfA, 10/2023
- Galaxy in the epoch of reionization in alternative dark matter UCLA DM meeting, 03/2023
- Alternative dark matter in galaxy formation Dark Cosmos seminar, Princeton, 10/2022
- Alternative dark matter in galaxy formation Brown Bag lunch talk, MIT, 10/2022
- Alternative dark matter in galaxy formation Harvard CfA, 10/2022
- Alternative dark matter in galaxy formation FIRE seminar, 10/2022
- Dwarf galaxies in dissipative dark matter GalFRESKA workshop, 09/2022
- High-redshift predictions from the Illustris-TNG simulations MIT, 08/2019
- High-redshift predictions from the Illustris-TNG simulations Caltech, 06/2019

SELECTED PRESS COVERAGE

[Kavli Foundation News: "Early" Dark Energy Could Explain Anomalous Galaxies and the Universe's Disjointed Expansion Rate](#)
[Univ. of Cambridge News: Dark energy in the early universe could solve two major problems in cosmology](#)
[UT Austin News: Early Dark Energy Could Resolve Cosmology's Two Biggest Puzzles](#)
[Yahoo News: How dark energy could relieve 'Hubble tension' and galaxy headaches](#)
[Phys.org: Early dark energy could resolve cosmology's two biggest puzzles](#)
[MirageNews: Early Dark Energy May Solve Cosmology's Top Two Mysteries](#)
[MIT News: Early dark energy could resolve cosmology's two biggest puzzles](#)
[CCA News: Bursts of Star Formation Explain Mysterious Brightness at Cosmic Dawn](#)
[Space.com: Dark matter atoms may form shadowy galaxies with rapid star formation](#)
[Northwestern News: Bursts of star formation explain mysterious brightness at cosmic dawn](#)
[Phys.org: Dark matter can make dark atoms, say theoretical astrophysicists](#)
[Scientific American: JWST's Glimpses of Early Galaxies Could Shed Light on Dark Matter](#)
[Astrobites: Forecasting the obscured first few billion years](#)